

ActInf Livestream #022.1 ~ Psychophysical identity and free energy

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okay hello and welcome everyone to actin flap this is the actin flab live stream and i'm pretty sure we're live yes we are it's number 22.1 on may 18 2021 and this is going to be really great discussion so i'm looking forward to it and thanks again everyone for joining live welcome to actin flab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are here on this page this is recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good live stream etiquette welcome scott and raising our hands to ask questions and hear from everybody because this is really going to be a special discussion we're here in paper number 22 and on may 18th today and next week on the 25th we're going to be having two discussions on the paper psychophysical identity and free energy by alex kiefer so thanks a lot alex for joining us live and engaging with the dot zero video because we gained a lot as a working through that and so we're ready to take it to the next level through conversation with you thank you thank you and today we're gonna be learning and discussing the papers link is here and the video is just us unpacking and exploring raising questions and we're going to be also taking notes so that in the dot 2 we can be following up we're going to start with some introductions just to introduce ourselves and also maybe to raise any thoughts or questions that we had at the outset about the paper so everyone can feel free to write down points or write down questions so they don't uh forget them and we're pretty much just gonna jump around to any topic or question that drives us along and we'll flow with it as people join and leave and there will be a follow-up discussion 22 next week that brings us to the introductions and the warm-ups so i'll start and we'll pass it around and then end up with alex who will provide i think a short introduction to the paper and then we'll continue the discussion from there so i'm daniel i'm a postdoc in california and something i'm excited about today was something we were just speaking about right before was learning about a transparent code and understanding what that

might mean i'll pass it to scott hello i'm scott david i'm at the university of washington applied physics where i run a program called the information risk and synthetic intelligence research initiative and i'm delighted to be here today and i'll pass it to stephen hello i'm stephen i'm based in toronto i'm doing a practice-based phd at canterbury christchurch and um i'm really excited about the psychophysical areas being brought in with this process with energy so i'm intrigued because that's connects to some of the research i do with communities in their landscape and how to understand that so thank you and i'm going to pass it over to alex thanks hi everyone i'm alex watkins i'm a researcher in systems management school in moscow russia and also co-organizer of acting club and i pass it to dean hi i'm dean i'm up here in calgary canada i'm retired and i just try to keep up with everybody here i'll pass it to our special guest today alex hey dave um no dave is left sorry okay continue okay i'll excuse me hey thanks uh yeah i'm alex kiefer um i'm a researcher in philosophy and cognitive science and i do some modeling as well um i'm i have an affiliation with monash university um i'm currently also a lot of my time is devoted to working for uh nested minds solutions which is um an enterprise uh started by maxwell ramstad and some other people uh just recently uh we've already got a lot of work under our belt um so i'm really happy to be here um thanks thanks for having me this is like i think gonna be a really fun discussion um cool maybe just while the other participants are preparing a thought or a question to lead off alex just how did you come to writing this single author paper what yeah in your path led you to write it this way yeah sure um so uh so this is an idea that i think first sort of uh came up for me in 2015 or so um so i've been fascinated by this formal analogy between optimization and uh processes and thermodynamics and physics um for a long time since well not that long maybe 2012 2013 when i started getting into the neural net literature on helmholtz machines and things like that and um so i think this this analogy uh between physics and optimization has been tempting for a lot of people to want to collapse these right and this is something that people like um my friend mel andrews constantly warned against is like just taking the physical informational description of the system to be uh the same as as its physical description but i thought that there was a compelling argument to be made here so i wanted to actually articulate not necessarily endorse right i think the way i couch things in the paper is a bit cautious on this but i wanted to articulate what is the most sort of the simplest most kind of perhaps radical perspective one could take on this the relationship between the cognitive dynamics and the physical dynamics um that you might be able to defend right from a sort of free energy principle-ish perspective um so i wanted to to um just lay that that one way of understanding

this relationship out that was that was as clear as possible um i think in the process i ended up writing a paper that many people find to be quite complicated so this was supposed to be an exposition of a simple idea so maybe we can see if we can get back to that awesome so anyone can raise their hand and we'll just start but i really liked how you said articulate but not endorse you've put out an artifact for us to interact with and to explore and you have framed the psychophysical identity thesis as a hypothesis so it's something that evidence can now be used to support or to disprove and that's sort of um a nice piece of work that now obviously we have started the discussion around so we'll go to stephen and then scott and then anyone else with a raised hand one thing i like about what you're doing here is you're sort of bringing it into the domains where the action is happening it's a bit like most of the physics around thermodynamics is really often used in chemistry and a lot like niels bohr was actually a chemist wasn't a physicist per se he used physics so there's an interesting thing about how these areas get transposed over you know may be layered again to biochemistry which maybe is trying to look at the mechanisms that sit on that and xyz so i think that's quite relevant to some of the questions we've been wrestling with with what happens when things start to transit transition to the ontology like and how how can you keep a handle on that when you've got people from different backgrounds so i i think that's quite interesting awesome thank you stephen scott and then anyone else with a raised hand yeah i'd like to just ask about your having written the paper and now contemplated it discussed it with folks what's your impression of the role of the observer of difference of gaps of differences in identifying sources of free energy um exploitable sources let me let me back up one i did a paper in uh presentation in mit in 2013 called entropy and i asserted that carnot's equation the hot and cold differentials that are reflected there are the same as arbitrage equations in markets because you and i asserted that heat engines were produced by the human mind observing an attribute because it didn't matter that we thought we had a flow of caloric and it was flowing downhill we actually the notion of flow was what we asserted in order to convert random heat random motion of particles which we really only use for warmth being able to put that into a cylinder a combustion chamber and then perform work took thousands of years i wonder if you could comment on it feels like we're using information like the cavemen as for warmth and not yet putting it in a combustion chamber and the combustion is it the fact that the processes where differentials are captured exploited utilized i'm not sure what word to use here is that a night like a reaction chamber are we reacting observations and data into inference in these reaction chambers of mind similar to the differentials that we exploit to in a heat engine for work are

we able to capture what are the devices of capture sorry for the long question right yeah your hand on your chin makes me think that maybe it's an intriguing question it's definitely intriguing i probably can't do justice to it but um i'll try um so just on the last point i guess i think that the um well actually so first to take your your question about information um whether we're using it um so are you sort of like whether we're really exploiting it up to its maximum possibilities right as information or just or just treating it as sort of you know uh something like warmth a source of warmth i mean i i take it that insofar as we're using so i right so it gets to the heart of my perspective and to this perspective of this paper i guess so i said i flowed this thesis as a possibility but i i do kind of have some sort of deep-seated conviction that that information can always be looked at sort of in terms of what it represents and and physically right so any process of information uh information is literally a process of physical change as well as as well as a process of some system coming to know something right so to take the etymology right just go back to shannon's original framing information is just the amount by which someone or something has been informed in my in my view so um so i think insofar as we're treating something as we are using it as information um right it's not now i don't know what the announce maybe i'd love to hear more from you on this i don't know what the analogy would be to using information just as heat is it like we're aware of something but we're not really doing anything with it like in terms of inference or uh so so i guess my my view is that we are in so far as we're thinking and exploiting information sources for inference we're using it to its um maybe not maximally efficiently always but we're using it in a way that's that's intelligent but but just one more point in this like um the the analogy to combustion engine and like the the idea of um i guess my thesis here is that anytime we're performing an inference any system is performing an inference the physical aspect of that is something like right it is an actual physical reaction so there is always something that has precisely that nature behind any uh use of information um but i think everything i just said sounds to me completely trivial as i'm saying it so that's its power yes perhaps what did you mean we can unpack that because it's known that is um not just simply wasted but where there's a heat differential there can be work and potentially when there's an information differential there can be an analogy to physical work but in the informational regime so what motivates that line of thinking for you yeah it was well and again i'll send around the slides from 2013 but but the idea was it is not by analogy it really is a capturing by the mind of an attribute of a system right you look at a block of iron it has spin it has temperature it has color there's all sorts of dimensions that it has that are unexploited um and then we can look at them

and exploit them for different you know magnetism with spin and then temper or electronic electrons flow through iron same block iron it's just sitting there but you measure different attributes and as you measure different attributes you measure them into existence and that existence then can be exploited and so we did that with heat right they thought that heat was caloric they actually thought it was a fluid right and that it but then the the guy who is the cannon borer i forget which one the guy who board the brass cannons he said wait a second it can't be a fluid because it's unlimited it's a quality of the material so i think what we're talking about in information it's a quality not of the brain of the it's quality of the mind and the differential between the brain and the individual and the mind there's a there's a mind out there to be learned let's call it an environment if it's animals but if it's humans it also involves language and when you're born you don't know it and then you go learn it so your brain has to get tuned into the mind so to me the mind is our combustion chamber and you gotta learn the combustion now if you do if you're different learn different things if i'm a hunter gatherer i'm going to learn different things in my environment to combust right i'm going to have different cues and i'm going to say wow this is relevant for me because i got to do this and this so you the we train i think we train our combustion chamber it's symbolic right since we're training this thing in which to contain the differential accomplishment of the active inference the beijing accomplishment and then we trained so it was an article in just the last thing i'll shut up for a minute there was an article in nature magazine a couple years ago they said less than 12 month old human infants learn in a beijing way if you point to it if you tell a baby a horse has four legs the baby will point to a table a coffee table and say horsey then you say no scott just disappeared it was too much of a truth drop with the table and the legs so stephen to you and we'll return to scott when we can okay please sorry okay scott's back anyway continue with okay continue from the table okay yeah anyways bayesian you they beijing accumulation you they'll point to a table then you point say you know let's have hair they'll point to a dog and say horsey then they say no it has to have a main they'll point to a horse and say garcia so it's a beijing the article was about bayesian assembly of human knowledge and so what if the baby is not really assembling the knowledge but just tuning their brain into the mind and the mind's and the mind we teach we have to learn the mind because that's the useful combustion we've already designed this community mind to navigate space and we teach it to our offspring in order to make them more fit anyway else cool thank you any thoughts on that alex well it's interesting i i don't know if i completely understand the way scott's using the term mind here but i think i kind of can see how something that deserves to be called the mind would emerge

through this sort of learning process right so so i guess there's a trivial sense i think in which anything that has energetic properties has a mental side that's kind of what i'm saying in the paper um but it's trivial and if you want something that really functions cognitively then you need to have internalized a lot of these regularities through you know the sorts of learning and infant schemes that we're familiar with here so um if that's what you mean by learning the mind i think scott you now guys analogized it to the environment at some point or maybe it's more than an analogy um it's like the the data the the source right of the data sensory data that you're interpreting um so uh yeah i think it's a cool idea cool stephen welcome blue steven with a question and then anyone else with a question or blue you can say hello yeah one question i'd be interested in is um do you see maybe more than one um type of information processing that may be happening at different levels so maybe you've got because we've had people talking say at the quantum level which could be happening particularly say the photons coming in with the iris and things going up and then you've got other levels where you've got these steady states going at different rates and then you've got some of the things you're talking about where maybe it goes into this sense of a more averaged closed mind idea which would have to be somehow created maybe because the dynamical nature of the data and the noise is not necessarily exactly how it is when it first arrives at the organism but maybe you're you're bringing back some of that more averaged out heat ideas um around how energy can be averaged over time in in some sort of heat form is that would that be accurate um i'm not sure about the heat piece i'm just genuinely not sure we can talk more about it but like as far as multiple types of information processing i mean i think that's definitely what's happening in the you know in the brain of course and there's all this literature on on multi-scale active inference and stuff which my paper isn't directly about active inference but um uh i think like there's no so we tend to signal out certain processes in the brain as like processing information but really that's going on on you know uh in on fine grain levels uh and on larger scales as you say um i think we tend to just pick out one piece to analyze at a time i guess the part of the question i don't know how to answer is like what what are the types of information processing in question so um i guess i'm skeptical of that type distinction not not against it i'm just not sure what what kind of types we have in mind but um definitely i think there's a there's a rule for averaging uh just in if we even if we just stick to neurons and don't worry about underlying chemistry or physics like uh if it can't just be that one neuron inspiring uh represents you know your grandmother because then if that neuron's shot then you can't represent your grandmother so um i think there has to be a if we want to associate content with

brain states for example those states are going to have to be couched at the level of averages in some in some sense or some kind of statistical uh property um and i i think that comes out clearly also in the work that carl firsten and uh banyan uh and others have done on on the whole markovian modism idea so they talk about the fact that it's um uh you know it's um it's not individual brain states that encode information on their view it's it's the the mean of the distribution uh conditional distribution anyway going a bit into the weeds on this but i i i do think that um that's important uh heat i don't know let's talk more about it yeah keith we had the what is temperature slide which i'll just switch to yeah yeah wait stephen do you have any other thoughts well you know don't slide and maybe just as just a sort of following from that is maybe something about the mind and the level of the psychophysical maybe actually the mind is kind of a psychophysical level as much we think of as a brain level but maybe what you're pointing to is something a bit more of an of a dynamic um which is kind of interesting yeah i mean i i generally sorry i generally have trouble seeing distinctions between levels so so maybe that's what you know you you all could help point out like why these things are not the same uh and it wouldn't would help me go ahead hence the psychophysical identity thesis paper that you wrote right because that's the entire topic of this discussion is what are those relationships and are they like each other or are they overlapping are they isomorphic are they one of the same so um here on this temperature slide you wrote about how increases and decreases in temperature change the variance of distributions so maybe just what is temperature or in this psychophysical identity mapping what does temperature reflect because changing the temperature on the brain you're not just talking about going into a warmer room you're talking about something related to inference so what is happening there well right i mean i guess i am saying like look if if you change the temperature in the room and that ends up changing the temperature of the brain then it's kind of an implication of the thesis that you are changing the temperature of your statistical model as well to some degree right probably usually to an imperceptible or insignificant degree when something like walking into a different room because now the brain has its own internal homeostasis and it's not like we're gonna drastically directly affect the brain's temp internal temperature um by changing the room temperature one degree but i really am uh you know speculating or i'm considering the consequences of i mean the overall theme here and temperature is a special case but the overall theme is that uh anything that you want like so any of the terms in the home holds free energy from from physics from thermodynamics um should directly this is a transparent encoding idea it should directly sort of be reflected in the cognitive dynamics and so the

cognitively speaking what temperature means like it says here is just um uh you know temperature parameters in models neural network models and other sorts of models um i think i've heard this referred to in the active inference literature as the as the shaky hand parameter right so it's like you're reaching for something if you increase the temperature parameter there's less certainty and exactly where the motor system will end up right so that's that's what it means cognitively speaking uh or if you don't want to call motor control low level motor control cognitive then in terms of of these models of organisms what it means is just literally just the variability of the distribution right the variance so um i guess what i'm saying is yes i think if this thesis is true then what encodes that is the actual temperature of the system that implements the model very interesting thank you we'll go to blue and then good morning i'm blue knight i'm a research consultant based out of new mexico and i really enjoyed this paper i really enjoyed the possibility of having a way to separate the mind from simulations of the mind i think that that's a really um relevant and pertinent idea to put forward but so in in relevance to temperature and this is going to like totally throw what i just said out the window uh you know i i um it dabbled a little bit in retail investing right and so like the day before or like maybe in the morning you might like what's called take the temperature of the market right so you see kind of as things going up or things going down like you just kind of put your feelers out there and take the temperature of what the market's going to do so i just wonder in terms of cognition in terms of active inference if there's a way to like take the temperature in in maybe like more of a um of a metaphorical sense right does that make sense um i'm not sure um so just i can elaborate a little bit more like what uh like in in the political system right we get like a temperature like is this like a liberal environment or is it like a more conservative environment and that might influence our actions right like the active inference loop might be influenced by the perception of temperature in the environment right so yeah i mean uh first of all hi thanks for joining um you you and dean and daniel uh you all i really enjoyed your discussion but in point zero and it made me excited to talk about it today because it's it's cool how just how many different questions or you know anyway whatever it's cool so um i mean sure you could use temperature in a metaphorical way as well like um i think uh it like i i guess in this paper i tend i i like i tend to assume that when people that that metaphors are are not metaphors but identities right as they graph the ground level assumption and then see see where that you're forced to distinguish between them so like if we're talking about again if you're talking about a model of motor control where you're literally reaching for something then i think in that case it's plausible that there's some relationship between the actual

temperature of the pieces of the system that are driving the motion and the temperature parameter in the model but like if you're talking about something like taking the temperature of the market or you know the political there still might be partial mappings there right like but you know if you're defining some arbitrary dimensions so if the the temperature politically means one side of the scale is is conservative and the other side is liberal then then i wouldn't expect that there would be some physical component to that um but the same way that your paper was uh as you said just putting it out there not endorsing the psychophysical identity it's almost like an econophysical identity thesis would be asking whether the thermo-like dynamics of markets actually have some sort of mapping that's more than a metaphor but it's something that can be stated and blanketed or boxed off and then can be understood as being adequate in certain situations but maybe inadequate in other situations but when we talk about the market heating up people know what that means in a way where you know the market feeling happy doesn't really capture it heating up or cooling down so hands oops i'm not sure if i should try to share my screen i may just screw up so let's not do that but let me so i just pulled up my slides from 2013 and so i said under the first law of thermodynamics temperature has two roles first from the perspective of a party outside the system temperature tells whether two or more systems are in thermal equilibrium same temperature equals equilibrium okay from the perspective of a molecule in the system temperature tells about the distribution of various molecules in the system over the possible energy states right the higher temperature equals a broader distribution of molecules in different states and a higher boltzmann distribution so what i'm asserting in this context in the context of those earlier slides was analogously here are we looking within this system of the cognitive system and the dynamics of components within it or vis-a-vis an outside voice for that system i think that's a relevant question here because when you're talking about the first law in the in here we don't we have um this is um um um maxwell's demon is actually a person now sorry that obscure reference but maxwell's demon was a convention to plus with the second law of thermodynamics a gating function um and now we actually have the notion of what does a molecule in the system think in a sense is what i'm inviting here and this and so i think the assertion and the assertion certainly i'm making that context is we can capture either one of those dynamics and it's info dynamics you're not thermodynamics is what i call it but essentially first law tells you about interoperability is what my assertion is you can have equilibrium in a symbolic way i assert is the equivalence of interoperability symbolic and so that's when you can have a beating of the minds you can have shared consciousness society

shared language all that kind of stuff is my notion of where the first law takes you thanks that's really intriguing um i need to read your work on this please um yeah i mean not to give a too hand-wavy an answer but i think since we're dealing with multiple scales here we could talk about neurons performing active inference people doing it i think both of those sort of uses of temperature are relevant but exactly yeah um we'll get to stephen but maybe just a general question where does active inference come into play and how does it come into play oh yeah so um right so you you guys raised in the point zero that i didn't really talk about activations that much in fact i only referenced the literature on active entrance um so this is yeah it's interesting this paper isn't really about active inference it's not really directly about the fep either so thank you for having me despite that um but it's i mean obviously it's in the it's in the very core it's concerned with the same core issues right but um i don't want to take us on this if it's a tangent but the reason that i don't write about active infants that much is that i think from a high enough or meaning abstract enough point of view um you you don't need i don't know that there's really a difference to me active influence kind of relates to the stuff the way that like dawkins's extended phenotype proposal relates to more traditional ways of thinking about um evolution and genetics in that it's a new paradigm but i don't i don't think that there's a system that you can describe using active inference that you couldn't also describe um using say traditional representationalist terminology so that's obviously controversial we could talk about it but um like in so far as your as you're concerned with analyzing or designing a system like right like if you wanna even if you want to write an active inference agent as a model you only have control over its internal states you can't directly control it needs yes it optimizes um its situation by acting and intervening on the environment but you can't directly control how the intervention on the environment plays out you can only control the expectations for sensory data given that you've acted in such a way and so that's why i tend to just default to you know it's what jerry fodor called methodological solipsism right which is a very different perspective from what people i think i'd love to talk about this maybe maybe they're more similar than i than i think too but um anyway so so uh like active inference is great i think in fact it leads to new ideas for technology and new models just because it's a different way of thinking about things but when you're doing like philosophical analysis like this i don't find that it's for me often necessary to think in those terms but i would love for someone to point out something that i'm missing by not doing well what we love here in the active lab is hearing everyone's perspective on active inference and also adjacencies and synonyms and other languages and other ways of expressing it so thanks for sharing it the way you did um steven if

you wanted to okay yeah so so building on that you you mentioned the helmholtz free energy rather than the gibbs free energy so you sort of the broader sort of energy that's available um the the i suppose a question i've got here is in in this in that um actually go back to that diagram just one sec with the distributions there you've got just you you've got both the difference in the sort of the size the distribution for the different size molecules but you've got the average speed is higher for um hydrogen but yeah hydrogen or oxygen it's it's faster it's the speed is faster for something that's lighter and that may have an impact at the local level when the actual atoms hitting each other at the same temperature but when you take the average it's all around the same energy if that makes sense so i just wondered if there's any particular sort of relationship to the way that the speeds are happening and the way that might change something at a certain scale and just helm holds and gives free energy yeah um so to be honest i mean the reason i wrote i write about helmholtz free energy is just because it's the thing that jeff hinton and others um used in their um when setting up things like the helmholtz machine um the gibbs free uh i'm just less familiar with it's it's it adds complications that i thought weren't relevant but if they are relevant then let's talk about them um yeah could we maybe hear a little bit from you just what is free energy in the senses that you used it or how would you describe it to somebody who might be not very familiar with thermodynamics right um someone such as me for example so what i would say is well i mean i'm a philosopher but um uh so free energy in a physical and thermodynamic sense as i understand it again welcome i welcome any uh kings in the other direction um but as i understand it it's uh just the energy that's available for doing work in this the system so it's the internal energy the all the energy that the system harbors just in virtue of its configuration it's sort of existence at that moment um minus all all of the energy that's uh not usable because uh it's it's tied up with with sort of this disorder um with with entropy um so i'm sure that someone else here could maybe articulate that more gracefully than i just did but that's that's the one term and the other term in the in the equation here is the variational free which is just um the difference between a uh the the basically the the gen the gener the generative model or the the joint distribution that defines the system's generative model of um some kind of statistical model right more generally and um the approximation to it that you've arrived at at a particular time step right so if you're doing variational inference you're trying to fit this q distribution uh which i mean you know people this has been talked about ad nauseum so it's hard for me to get the energy to really launch into it because i feel like we are familiar with this concept but it's it's just the difference between the approximating distribution the recognition distribution or density and

the degenerative density so so it's so i guess the thing to underline there is that it's it's only defined in terms of these p's and q's right q recognition p generative that are they're defined in terms of the generative model which is a model right it's a it's a it's a um a mapping between like hypotheses right or hidden states outcomes so there's obviously no a priori connection between that and like thermodynamics um so it's just important to keep in mind that one of these is in terms of how it's defined as thoroughly like cognitive has to do with belief updating and the other has to do with the physical states of the system and that mapping between the cognitive and the inferential and the physical is again the entire discussion and the entire point of putting out an artifact that is critiquable or is extendable in the ways that you made it so scott then stephen and then also anyone else who's watching live can just put a live chat in and we'll ask the question so and then stephen so this is really what a great conversation thank you we've got a great paper so couple things first first of all you know i wonder we always talk about it being by analogy and it's not really thermodynamics but von neumann said to shannon use the second law of thermodynamics for your quantitative theory of information because in his quote right his quote was because the math is identical which tells you something it at least tells you something about math and humans but may not tell you anything about the environment but it tells you something about math and humans but he also said and if you use the word entropy in any argument you'll win every time because no one will know what the hell you're talking about so that's good advice on both of those points right so i always look at it and i go okay so let's talk about those differentials and you know we took that free energy question that we just answered i so i deal in policy i'm not a biologist i'm a lawyer okay in a physics lab and the reason i'm interested in this stuff is i think you can access the external states and in fact i think that's all we do as social beings is we that's what rhetoric is i am constantly trying to lasso external states of everyone else that's what politics is that's all the whole misinformation problem right now we're setting out now in a biological example we kind of don't send around memos as much now there is there are communications obviously in in the sciences and in the nature and all that but when we get into the human populations we may have something more going on here maybe mind is an emergent phenomenon merely of the attempts at harmonically coupling with other minds maybe a break if we're successful in tuning into that local mind we survive if i am dropped into france and i don't speak french then am i going to survive now now i will but what if it was 300 years ago or whatever right so you need to if it's if the mind is fair feral and is trained maybe that ability to affect the externality is everything maybe that's how we are humans in the way that we are because we're pretty

pathetic little organisms on our own each one of us right we don't have sharp fingernails and our tooth and claw is not would get bloody but it would take a lot of biting because they're kind of blunt right so anyway that's a bit of a tangent but so the question is are we maybe if we look at the maybe we're not entering another brain maybe the mind is where the action is and when we're doing this in training maybe we're merely tuning into something that exists already in our in our attempts here at this for active inference and we're just getting on board with the existing mind and so therefore our steady state our homeostatic state is the one that is most aligned with the externality that's drawing us there and as lawyers and as policy people we create artificial externalities that's why i use synthetic intelligence in our title we're creating synthetic externalities and saying hey come on over here we've got a good externality over here check it out this one's resilient sustainable and the sdgs in the u.n love it right so we're creating realities what do you think about that in terms of the notion of the uh efficacy of the possibility of harmonic coupling with other minds and that as a another active activity in active energy is not just changing your internal model yes it's extremely interesting so um i mean i think this question of how exactly the external states get into the picture is is like pretty fundamental for a lot of people who are in this area and uh i i think you're right that it's it's it's essential but it's essential in a strange way where it's completely inessential too so like if it's it's essential in that the yeah the reason that your steady state is what it is essentially involves the environment right so like in the free energy principle the the nest the non-equilibrium steady state is defined with respect to the situ this you know a sort of a subsystem that's embedded in an environment and that steady state wouldn't exist without both pieces of that right and it's the fact that you're doing gradient descent on that that guarantees that your variational inferences is going anywhere useful uh and and that's that's sort of the sort of you know magical seeming connection between actual accuracy and and all this just churning that happens when you're minimizing free energy over time um but but the the way in which it's inessential is of course i mean i i just i guess maybe i'm maybe i've been called a cartesian before maybe i am i don't know i didn't mean to be but um there's a sense in which simply it just the datum here is just that i can get things wrong and i don't see that there's any obvious way of limiting the scope of that right so just fallablism um so i think there has to be some uh subset of states of the overall universe that you'd identify with you your perspective and otherwise right you have sure the external states are essential to this story and they drive the dynamics de facto but there has to be a difference between the actual external state at any moment and the external state that i represent by my hidden states or my my internal states um

and i think you know like the reason any of this works in prac in practice is because we've internalized the structure of the external environment and this whole point about harmonic coupling between minds is really cool um it's there's something really cool and bootstrappy about that it's like right minds are things that harmonize with other minds right and um i don't just to kind of pick out another piece of what you said i um the way that i tend to think of this stuff is in terms of sort of imitation so the fundamental term for a lot of people in this area is prediction but i almost think it's more like we just learned to our brains learn to imitate the the sensory uh right the patterns of the processes that produce patterns of sensory input but the reason it's not just imitation is that also the world is just made up of a bunch of these generative models that each have their own driving right um signals so um so that i'm not sure i don't know i don't know how to tie that up in a nice package but it's it's a cool set of questions um and uh but so so the reason though that i stopped short of saying that the external states are as fundamental in understanding cis systems biological systems uh people etc as the internal states is simply things like cartesian skepticism so we could talk about that um well yeah go ahead scott just a follow-up well it is interesting because it's um because our struggles in it when we you know i say on this this group very often the only time you see reality is when you see paradox right if you're seeing absence of paradox you're intrinsically seeing a model of some sort and so we should embrace a confused state and and know that that is a reality and so that is not just a statement for this conversation but in general maybe what the mind is doing is the same thing that the body does when it's standing still you're actually never still you're always dynamically moving to maintain a semblance of stasis in a banded stillness right so maybe we have a banded mind maybe it's fully dynamic and maybe what we think of as us an existential kind of element of it is within a band right do people change over time it's a ship theseus in a way right you're still the same person but you're some of the stuff has changed out swapped out over time um so maybe the embodiment helps us with mind as a fractal way at any level because we can say okay if we have a system that's doing processing of dynamic environment some of that processing may be internal some may be external i mean that's group formation and you know part of its genetic and consanguinous in terms of performance and some of it is people who are similarly situated right if you're in a disaster and everyone's all of a sudden has the same problems people band together all of a sudden very easily and their mind morphs into that organism that is needed for that acute situation in that stressful environment and so they maybe it's always dynamic but that we don't see the dynamics as much and they seem more leisurely when we're studying them in the abstract but maybe

they're highly dynamic in both directions especially as we become information beings and more dependent on symbols thanks scott just one thought on the dynamics is anyone who's recorded a neuron or thought about that knows that just to keep an idea in your mind for a few seconds is being underwritten by vast numbers of dynamical processes that are quite rapid and the second point about the internalities and the externalities and the internal states and external states the helmholtz free energy has this u internal energy and that's like the configuration of the molecule usually thought of as and then this negative ts is sort of that those vibratory modes that can't be as easily harnessed but when you pointed to synthetic externalities which is the same thing as saying reconstructing the internalities it's like what if all of a sudden the organization of the system we could partition it a little bit differently so that we had more u and a little bit less ts and when you think about a vibrating molecule u makes it look like the covalent bonds and ts is the vibration but if you think about an ant colony the movement of the pieces is still part of the internal dynamics and so especially when we're talking about information which isn't covalently bonded atoms internal energy is something about the model it's not just about how things are touching and can be broken and reformed so there's a lot of awesome pieces in what you added there steven and then anyone else with a raised hand yeah i think building on that you're right there's this in this i think you're making the right choice here for this paper using the helmholtz free energy i think that's a um even if gibbs free energies use more in active inference that adds an extra level of complication and i think this is actually useful then because you're able to look at some of these distribution questions and just how information is extracted but it's actually helpful for me to see you go back to the helmholtz because it makes me realize well that gives free energy which is more about how action happens how do you know what the internal energy change would have to do if it was trying to also keep within a certain constraint of volume and pressure which you don't have to worry about with helmholtz free energy that would bring in action that would bring in the action part of active influence in some ways um but the underlying principles still hold for what you've got you know so i think that's um that's helpful so i don't know if you've got any thoughts on that yeah yeah that is health i would just i would just say that that is helpful and um uh i would love if in the future anyone yeah worked on this same topic and one and generally you know ex unpacked the the sort of um i was going to say generalization but it's maybe the opposite um the the how this would work out in the case of of action more specifically and using gibbs free energy and stuff that would be really cool awesome and taking the temperature of a person in terms of going for a checkup to the doctor and they put

the thermometer under your tongue and that's a thermal temperature and how are you feeling what does it look like to take the temperature just blue had been suggesting so maybe let's go to the transparent code um slide that you added alex because that was definitely something that we were curious about from the dot zero and we can just hear your thoughts on this talk a little bit about this and then also anyone else can just write down a question raise their hand or ask in the live chat but we'd be really curious to hear what you thought about this yeah sure so um yeah so the reason that again that i put this in was that dean and um and maybe maybe everybody in that point zero um discussion had um had questions about this and i think uh the transparent code it ends up sounding like a slogan that invites you know speculation and it's like it's a mystery or something but um this might be a case where i'm trying to say something so simple that it's it's easy to miss right so so all um the role let me just first frame this the role that this plays in the paper is that if you want it to be that variational free energy and hellhole's free energy in the thermodynamic sense um are the same quantity then you're going to have to um explain how well both of them are sort of made up of probabilities right so you need to have a mapping from one set of probabilities to the other um so uh there has to be a way of understanding how the probabilities of states this is a big distinction in the literature right how the probabilities of states could represent could represent so the probabilities of states in the system the cognitive system that you're analyzing which is also a physical system how is it that those could represent um the probabilities that are assigned by that system to the external states right um so so the idea of a transparent uh neuronal code or any code uh is is just that the probability of for example so in the on the left hand of the diagram here if we had a transparent code in the sense in which i'm using that term then for example the probability of neuron b firing given that neuron a fire is right so if b is postsynaptic neuron and a sends a signal to b or a a fires and uh there's a certain probability of b firing subsequent to a firing that would represent uh the probability of some state of affairs q obtaining given that some state of affairs p obtains where a and this we can we can talk where this mapping comes from right but the assumption is a represents p occurring and b represents q occurring um so so that's really all i meant by a transparent code and it's it's not it's a simple idea but it's it's definitely controversial right like that that would mean that if you think that there's a certain conditional probability of you know the cat being on the mat given that the cat lives in the apartment um then sort of the probability that you would represent those things has to be tied up with the probability that you assign uh to those things occurring and that's not at all a sort of um uncontroversial statement um but it's a simple idea and this is an idea by the way again that's

that's used in some of the early connectionist literature um on um blue mentioned um boltzmann machines right so there's an early paper on both both of the machines uh where this kind of stochastic code is proposed and the basic one more sentence on this the basic idea is that the objective probability sort of measurable probability of events within the system uh maps onto the subjective probability that the system assigns to the states of affairs that are represented by those internal states interesting and in the context of the cognitive system of interest we're talking about the brain it's like we know that we're not going to get the transparent coding from the photons hitting the retina to what is the probability of the cat being on the mat so with this multi-level processing as you've kind of hinted at with the multiple layers here it could be possible that at some neuronal stage or at some level of processing there is a pattern that maybe represents the cat being on the mat probabilistically which also invoked the whole representation questions which we had talked about months ago um right so it's so fun to see how you you delineated where there's controversy sometimes even both sides of the discussion around a controversy and then it's like great here's the artifact let's see which way we're gonna push it if there turns out to be more evidence for transparent coding then we'll push it that way and if there turns out to be less then we can push it a different way so um scott uh if you're there i don't see you yet so stephen first and then okay stop scott go ahead and then stephen um i just wanted to mention there was an article in nature a couple of years ago where there was an assertion made that consciousness this is what caught my eye consciousness they asserted arose 520 million years ago and they and i was like well that's interesting what's the evidence and what they found was some structures for i think they were proto-nodo cords or something but basically it was a ganglia kind of thing a secondary processing and their assertion was that because they found physical evidence of a neuronal again in in fossils it was some eddie karen are able to be preserved in shale or something but um uh that because it was secondary processing beyond the the eyeball or the ear or the perception apparatus that secondary processing they suggested meant there was consciousness because that was their definition but it was interesting the reason i raised it here is it's interesting that notion of embodiment right so we're talking again let's talk about reaction chambers we have there's stimuli all over the world there's things happening and then we have perceptions ability to detect stimuli right in our afford it to us and so that comes in but we don't have the structure of it is tbd right and so then we the the the organism embodies morphologically certain structures over time eyeballs notochords and ganglia and things like that right and those themselves are information structures they just haven't even embodied in a

morphology so it's morphological computing for biological organism right because they now can do that thing i can't see an ultraviolet but bees can that kind of thing or infrared that kind of thing um so it's so i just wanted to raise that idea that assertion that offended me originally that consciousness arose 520 is not so offensive in a way because what they're trying to do is grope at the notion of the layering on of morphology and embodiments and say look there here was a ratchet here's something happened here and we think we don't know how much more experience it's like julian james when he went back and did the origin of consciousness and the evolution of the bicameral mind book in the 70s j-a-y-n-e-s and he was asserting from a literature search that the wernicke's area flashed on at one point it was equivalent to broca's area language center in the other side of the brain and his literature search he looked at joan of arc he looked at the iliad and he said that morphologically embodied that notion of the mind actually changing ancient literature can see the mind turning on that internal voice and so i just wanted to raise that notion those embodiments of the external and internal states lying around all over the place right it's your paper does a nice job of inviting the um treatment of those or those those as candidates for looking at the mind being assembled particularly if the mind is external to the brain and the brain is just an antenna tuned into the mind thanks one one funny little synchronicity there is that uh julian james wrote the book that scott mentioned origin of consciousness in the breakdown of the bicameral mind but edwin james was a very famous researcher who introduced the maximum entropy interpretation of thermodynamics so related like we all are bi were they related were they related by cons the blood aren't we all yeah oh come yeah just sort of coming to this diagram that you've got here um so i think um one thing i'd be interested in unless you say neuronal or other internal systemic states so you sort of give a little bit of latitude to dynamical aspects of that goes to cognitive states of the system beliefs and i'm wondering how much you're tied to those beliefs being um statistical computational sort of separate things or how much is beliefs on action policies i mean i know i'm pushing active information again but yeah it's you know it's fun yeah what's your thoughts on that yeah i mean there's all sorts of questions here about how we individuate representations and such but i wouldn't think i don't even i don't really think of these as detachable uh separate entities at all so that node is labeled p but i'm one of these functionalists who thinks that the reason that the label itself does nothing right the reason the the label is a useful indicator for people outside the system to understand what it represents but the representational function is just the role that that node plays in the network so um um and that would i think that it subsumes and includes um connections to action right i just tend to think of connections to action as the

special case where what you're generating what your generative model is generating are states of the motor effectors so right i just and i think any any kind of planning um that that that the organism learns to do can also be understood as learning sort of sensory motor contingencies which will influence how states of the motor effectors are generated in tandem with states of the perceptual system right how perceptual biases occur so um basically yeah i don't think that that there's any like these don't need to be cognitive separable features of the these don't need to be separable features of the system they're like when i say that a state represents p um because i'm represents for me has sort of a functionalist meaning i just mean that's just sort of a label for uh a set of uh ways in which the system can behave because of its internal structure so including sort of internal um dynamics right but it's really just a label for a property of the dynamics of the system cool and also beyond the action being at the gross organismal motor level potentially internal representations or states could be like hormonal or neurosecretory so the neuron can fire but neurons have other affordances like the little micro motor behavior with the motor proteins like releasing a vesicle of of a protein that's been transcribed and translated also tiny little motor behaviors of nested smaller agents inside of that neuronal interface so stephen and then anyone else with a raised hand actually just what you were saying there daniel does that mean that you can almost release hormones through the brain it doesn't have to be like through a gland in the body but you've got smaller scale releases neurosecretory cells and neuroendocrine biology are large areas so yes many kinds of cells pull in and secrete molecules and neurons are of various subtypes secreting specific hormones neuropeptides all kinds of hormones and signaling molecules short range and into the circulation okay thanks um and i let that process and that's really useful i didn't know about that and um i supposed just adding on to this you you when you mentioned about sensory motor contingencies and and and being able to um so i'm wondering that the scale of um consciousness you know and things entering into consciousness or the phenomenological sort of awareness do you sense that there may be even if active inference is under writing it all some other types of like it doesn't have to be as pure active inference at that scale and it could be working with other types of sort of architectures or process architectures uh i mean well one thing is that consciousness so i think scott brought up consciousness for the first time here that's really interesting i um i sort of deliberately didn't say anything about it in my paper because i find that it's um it's a very thorny topic for me of course for many people right it's not it's not notoriously not an easy problem or maybe it is and it looks like a hard problem but anyway um yeah so so i'm not sure i i do think that like all of the cognitive dynamics i'm

talking about are probably more fine-grained than what shows up in consciousness so i wouldn't say that there's some transparent mapping from whatever you believe in in the sense of belief that's implicitly defined here and what you are consciously aware um i so we can there's all sorts of theories of how consciousness relates to this stuff um there was a second piece to what you said that i wanted to address but i don't remember what it was well i suppose there's a scale but i think you kind of addressed it a little bit there sorry daniel you're gonna say yeah we'll we'll come back to it if you remember what it was yeah scott and then yeah i wanted to go back to the notion of neurotransmitters from a minute ago when you're talking about endocrine neural function a couple of things there's a couple of books here just want to hold up this um cognitive archaeological the cognitive archaeology and human evolution kind of an interesting notion there's a couple authors there here's another one archaeology and memory kind of an interesting one the reason i raise these is kind of interesting when you talk about neurotransmitters i remember reading an article a few years ago that suggested that among all the animals in the animal kingdom now the animal kingdom is an old representation corollas linnaeus thing i get it but in any event um there's only one animal that doesn't share any of the neurotransmitters or the rest of the animals like the box jellyfish or some kind of jellyfish i forget which one it was and i thought that was interesting just what we were just talking about a minute ago daniel was noting about neurotransmitters well again it's morphological computing those are artifacts of earlier solutions their practices of processing of information by these organisms right and so the fact that we can go back and say wow i wonder if these things were related given the fact that these don't share any neurotransmitters that's kind of an ethereal element it's not something that you get like bones and you know it doesn't fossilize well but isn't it interesting that we can find the morphology of neurotransmission in current organisms and use that to understand lineage and relationship so in fact that's a it's a morphological artifact and it's not epigenetic right but it's interesting the relationship of the the uh homeostatic state of uh thinking the cognition of a certain line of organisms and the morphology of its i'm interested in you social insects for instance do people do profiles if anybody here knows anything about you social insects please type in the um do they what's the story with neurotransmitter analysis among you social insects is that a special neurotransmitter specially invoked and is the morphology of neurotransmitters perhaps where the organism resides more than other physicalities thanks great great question scott so first on the tina four point and then the eu social so um the tina four indeed it did suggest new evolutionary hypotheses when people investigated which neurotransmitters tenoforms used

and they convergently utilized some of the same simple charged amino acid derivatives but they use them in different ways for example glutamate is not used in muscle contraction in vertebrates we have acetylcholinergic receptors at the muscle but glutamate was used which supports alex's point about functional representation there's nothing that says glutamate means muscle contraction it's a functional context with a receptor and a muscle cell and a neuron that generates the contraction and potentially for a later day maybe when we discuss active inference the paper but um we've done multiple studies measuring metabolism and gene expression in ants and there's been a lot of work on honeybees as well and just the one point i'll make here since it is a large area and people can look at the papers if they want is that the neurotransmitters that are involved in the regulation of foraging in ants and bees are inherited from the neurotransmitters that regulate foraging and solitary insects like dopamine and octopamine in fruit flies where it's been investigated in the context of solitary foraging but the inputs and the outputs of foraging for the eu social insects are different so whereas in drosophila it's like internal hunger signals representations maybe some sort of neural hormonal encoding leads to foraging behavior and then that results in the acquisition of food that re-enters that bodily but an ant forager is foraging essentially food for the larvae a different part of the colony organism so the inputs like the stimuli that lead to induced foraging behavior and the outputs like where the food goes are different but the foraging motor behavior is conserved and it's that rewiring of the motor motif of foraging via the same neurotransmitters octopamine and dopamine and others that actually enable the colony life cycle so it's a really awesome area i hope we can talk more about how that relates to so um stephen and then scott and then anyone else who raises their hands yeah i'd like to ask a question about which sort of links to that it's a it's a kind of a semi theory i've got is it feasible with this that the brain like an organism basically the cells that become the brain effectively they mutate and they they transform into the unique form that the the brain and the nervous system basically like another organism growing into the main organism right so you've got two organisms once they differentiate and their brain is like the control and the nervous system is like the control organization growing into and with the other the fascia of the other organism well in a human fascia but obviously i just wondered if that would be plausible and that that would do anything with this brain organism boundary question that i know are kind of connected to in terms of wanting to maybe maintain more of that potential i don't know if that's something that other people have talked about or whether it's a little bit out there and just um yeah biologically or from the sort of cognitive philosophical sense how that pans out yeah yeah i love i love the

like biology and biochemistry and stuff and all this you know wonderful fine grain scientific life sciences stuff that you guys are bringing to the table on this um that's that's not as much where i'm coming from but i think it's really really cool um but so so that steve the idea that you just articulated sounds sounds out there and plausible to me so um i think that like uh that's kind of what i would expect is that this same story can be told on multiple scales internal to the system right and so the brain learning sort of learning growing uh i guess i've just been conditioned to think of all this as one type of process right so i said learning but morphological development and growing into its into its um mature shape is a form of learning but so anyway um that happening and interfacing with the rest of the organism um i mean i definitely don't see any reason that we need to draw a like joint joint in nature around the physically right the organism that we can perceive there are going to be many similar joints that we could delineate within so um i think that's a cool idea i don't know much about the details of that process so i can't like check that validate it but it sounds plausible to me thanks alex blue and then scott so just to kind of maybe um ask steven to unpack a little bit what he said so i think um in terms of like a brain organism like separation or duality like do you maybe mean like mind organism because as like in my experience it's like the um like the brain is like the control center right like i don't know that there's necessarily like a division between the brain and the and the organism but definitely like the mind is kind of like this ethereal concept so maybe could you just unpack that a little bit well i was actually thinking partly thinking about the work of mike levin um mike levine or mike levin and um so in a way i my understanding is that once a cell becomes a neuronal cell if there's no going back like it's kind like it so it's in some ways it's um it's it's i was curious whether that whole structure be it from a small organism is effectively like we basically have two organisms in our body because our cells have differentiated into the body organisms and the neuronal ones and they've sort of they they work together in this symbiotic kind of cybernetic loop so that that's the kind of thinking um about how it pans out i'm you know so i just if i could just like speak to that a little bit um so the neurons can de-differentiate and do um in like especially like in neural cancer types so we do see some some d differentiation there it's not as committal as we've thought before i mean this it's kind of a budding field and the the you know antique way of thinking is that there is it's a set lineage but there's there's ways that you can convince them to do otherwise i think mike levin's thing is the bioelectric field right and so so he separates the the bioelectricity of the organism as as maybe the overarching control center there but i've not seen like a neural um phys like a neuron body distinction that's why i was just like what sorry well one thought

on that and also to scott's point about the emergence of a second level of the nervous system is to remember from developmental biology that ectoderm the outer layer of the developing embryo turns into neuroectoderm which is the tissue that gives rise to the nervous system and gastrulation in the neural tube so the nervous system developmentally neurons share an affinity and a developmental proximity to the external layer and so when you start nesting internal layers inside of that tube that's closed you start getting higher order nervous systems and then also the pituitary gland to sort of close this with a neuroendocrine discussion the pituitary gland is actually amalgamated from multiple developmental sources and so it's interesting that it has such a pivotal hormonal position integrating hormones and the brain's function and it itself is like a re-fusion from a tissue perspective of multiple branches of the ectoderm that had branched and then they reformulate a new organ which is the pituitary gland that serves a very coordinating role in organismal development so scott and then anyone else with the you know this is the best way i can spend my day with people who say the quote the pituitary gland is amalgamated from multiple ectoderm sources there i'm in the right place man you know when you you know you know this is the place for me i can tell you that hey um a couple things let's go back to this part in holland dynamic and cancer as an organism stuff for a second this is uh this is good stuff so you know the cancer analysis where they start to treat cancer as an organism in an ecosystem you know some of the framings to think of it that way a recruitment of a kind of chaotic kind of like having a drunk uncle stay at your house is one way of looking at cancer right it's a it's a chaotic use of resources um so that is kind of interesting right so we're talking and going back to steven's point about the mind being different so i'm thinking of symbiogenesis here lynn margulis stuff and the chloroplast in all plant cells and the mitochondria and all animal cells were once their artifacts of once pre-existing bacteria right they migrated on board with eukaryotes because it was a better deal right more leverage more de-risking higher levels of engagement with the outside right they change their internal physicality and morphology so they could think at way higher levels they could access way more complex levels of engagement than single cell organisms including the engagements they could get through sexual reproduction which introduced additional symbolic novelty so they could be more dynamically engaged with environments right clonal reproduction does not introduce a lot of novelty so it's hard to keep up with dynamic environments if you're formal this is something like intelligent design but it's not so it's interesting steven in your notion of the mind being separate because in a way the mind is formed by the social environment you know i've told the story before my sister adopting a

five-month-old chinese child who is now 22 who is raised in central pennsylvania speaks spanish and english does not speak chinese does not like chinese food does not like chinese music and basically looks like a chinese person but is not so is that mind a chinese mind the problem is we don't get have any control groups on the mind formation right once you're formed you are who you are and we say okay that's it but i'd like steven where you're going on that notion maybe the mind is an independent quantity and again the brain is an antenna to the mind so maybe the body contains the antenna components again i don't mean a real antenna but sort of um and it contained it needs to be tuned it's feral and so maybe what's happening is the tuning actually creates a mind maybe the mind is not resident in in a child and maybe bayesian inferences construct a mind and to maybe direct it to alex before we go to steven's question dean you really focused on this sentence it may be for example that biological run an algorithm closer to of divergence than to simulated annealing so what does mean or what what does that reflect on can i indeed maybe give a little extra info on that just just a quick quick thing i want i want to hear stephen and i want to hear everybody else but i have a question in the background sorry i have to say this in the background i've got people taking the front of my house off so i haven't been able to do much other than try to control my add but i have a big question for alex once we get through the the part that stephen and scott want to have addressing it's specifically to what you just raised so thank you great let's go to steven and then we'll return to dean and the contrast of divergence yeah maybe this will bring this together is i suppose what i'm just proposing is is that there is some structural biological differentiation which is will have an impact in some ways and how we think about extended cognition but if we i i'm very much of the mind body environment dynamical system kind of so i suppose i'm i'm talking about there being a difference biologically but probably at the macro level i'm i'm trying anyway to formulate approaches which talk about a whole and not anything else than that but i recognize that it may open up some other cans of worms um i i know that how i wonder if that's something that you're does this does this annealing relate to the mind-body environment dynamic in playing out within that or is it a little bit more separate or different to that yeah i mean the the use i was making of this contrast between contrast to divergence and simulated annealing here is was just was kind of more specific um i was just thinking of so annealing this process of of gradually lowering the temperature and simulated annealing was a optimization technique that mirrors that process and um so i mean i guess the answer to your question stephen is i i wasn't really thinking about system environment relations here at all i was just thinking of um two different sort of internal processes and how they might be

mapped on to um onto these these uh learning algorithms and processes and physics but so i mean i can wait to go on to that to take dean's question or yeah or whatever maybe let's uh let's go for the divergence yeah i i'm not predictable um i'm going to pull up the paper because i need to uh explain this so alex in the paper you you talked in in the subsection one two the role of internal energy um let me just find it here you talked about the inverse relationship between probability and is exploded and most uses of energy as a cost function and optimization problems so you made that clear and then a little further down in the paper you talked about contrastive divergence which in equilibrium distribution of a restricted boltzmann machine and then you got into a little bit more about the the comparison is the simulated annealing and then you got into section two and i'm just going to try to find it here the ensemble piece so here's my here's my question um if i were to bring a um if i already bring three things and and cast the widest largest wrist and blanket yet around your paper because i think that's what your paper does if i were to bring a polaroid picture a sprinkler and an astronaut together so if i were to bring identity which is what was the primary attractor of your paper to me identity astronaut that that distribution piece and um the first one the the the polaroid the isomorphism could you explain how you you brought these things that are so seemingly um distal into some sort of a coherence how how did when you were sitting down because i know you did this on a very proximal level but it it has far reaching implications based on the kinds of questions that stephen has asked the ones that scott's asked the points that lou and daniel have made it it doesn't just seem to be local it seems to be really distributed as well and so i'm just really curious you wrote the paper but now that you've kind of been able to step back from it and sort of take it into perspective that distance must have you really really looking at this maybe maybe in uh interesting ways because there isn't just the stuff on the paper there's the stuff now that you've kind of reflected on and seen the inversion and and wondered holy macro did i open up a can of worms or what did i what have i done i'm just i'm curious about that on a personal level right um well just i mean in general my the most striking thing about how this has played out um is that people are really interested um so blue mentioned this in the discussion i had emails about it too are really interested in the distinction between um genuine minds and like simulations um which was completely an afterthought when i wrote this it was just like oh yeah i better deal with the scope of this and so i should put a sentence about that and i did put care into it you know i i put the best sentence i could and but it wasn't my focus but um yeah the i mean the the the the thing i guess the thing that that strikes me about this so it was it right i mean it was intended to be this it was it was intended to be a very fundamental

like thesis that i'm considering here so i do think it has far reaching implications i have no idea whether it's true but right if it did it would have at least far-reaching implications i guess what um what i'm less clear on is well i i i'm still i'm still waiting for the other shoe to drop i'm waiting for someone to say here's an example of a biological system in which there's no way that you can map the cognitive interpretation directly onto the physical interpretation because um if that's the case and if i think if it's a compelling example then then i guess i'm wrong so far so far that hasn't happened there have been some interesting suggestions um but so so i guess i'm i don't know if i'm directly answering your question or dean i'm just um to me the the devil is still very much in the details and this is essentially a defense of a pop i was trying to i was trying to show that maybe some some a priory prima facie objections to this kind of thesis were not uh you know were not compelling but there could be all sorts of details that merge that that might question it um so i don't know if that really answers your question i'm sorry no no it and again it was kind of like when i was talking with blue and daniel in the point oh and i i brought up so i i talked to people who who very much are in an activist camp and you know they want it they want to see that realistic and then there's the statistical camp and they really want to have that sort of top down where's the channel what's going on and i i just kind of thought what your paper did is it it opened both of those perspectives i don't know that they even have to necessarily merge or overlap i think the fact that we can we can use both and see both it's just a really interesting thing now i don't know whether that is disprovable or it should be uh i guess that would mean that an activism is not a thing or statistical modeling is not a thing and yet we use these things so i i mean i guess they can be discounted but i don't know that they'll go away that's all i so i guess you have answered my question we can leave things open and up and not necessarily have them provide everything but but we can't just sort of discount something and throw it away until somebody as you said comes along and says no right it doesn't work yeah and just one quick follow-up on this this the one reason i wrote this paper is that i think what a lot of people find compelling about the free energy principle and the literature around that is the idea that there might be a link between physics and cognition right that there is a fundamental theory kind of emerging but it's i think it's very hard to pin down what the precise relationship it's hard for different people at different times it's becoming it's becoming clearer this has been worked out a little bit in the free energy principle literature but like it's it's difficult to articulate exactly what the the deep connection is between you know variational inference on the one hand and and like um the behavior of physical systems so i just wanted to i you know i wanted to throw out here's one

um version one possibility of what the uh supposed to be a clean precise brief statement of what that relationship could be um so that you know it yeah it's open-ended but so that it can be you know false falsified juxtaposed to other um other interpretations of that relation etc thanks alex we'll go to blue and then if anyone in the chat or if dave has a comment then feel free but first blue so alex i think that this goes back to what you were saying in the very beginning about scale right so it's it's it's hard right now even to map like cognition onto like neural activation i mean i mean right now we don't know um too much like too many components in are are murky for it to really be clear what's happening and i think you know as we talk about information going from a cell you know to from a brain to a neuron to cognition to action to bodily regulation and these kinds of things we don't really know is information compressed is it integrated like what is happening to scale a system up or down right i think that that's a hard problem yeah i mean maybe i'm not sure what you mean by scaling the system up or down um in in this section you talk about the scale-free nature of free energy dynamics and about how global and local free energy minimization could be understood as subsystems that undergoing some sort of uh akin process at their own skills right um yeah so i mean the the thing that so i mean the thing that i wanted to react to and what blue said was um the idea that we don't really know in the brain still it's very murky i mean i think that's true it's always relative right to some ideal of how much we could know but i what got me into this whole literature in the first place was was i was i started i starting again starting with the machine learning literature around 2012 2013 i started to feel like we actually were getting a grasp on on some of these principles how this works quite generally so like like the helmholtz machine for example um that blue you talked about that in the point zero um i think that's a model now if the model is accurate then it seems like what's happening is as we um you know as signals are sent from one neuronal population to another um further from the sensory motor periphery um the information is being compressed right so that's a particular hypothesis there's a successive compression of meaning that things are represented using uh you know lower dimensional um well vector spaces if you want to translate between vector spaces and neuronal populations right um and and that we have a pretty good right like this this sort of architecture explains so many things i mean that's this is this is what got people so excited about the you know predictive brain or the bayesian brain type hypotheses is that there's so many different things from different domains in psychology and neuroscience that this seems to this kind of perspective seems to unify you know it's it's it's to be expected in a sort of helmholtzian architecture that like you know uh the you can always sort of endogenous

endogenously cause the same sorts of states that you could be in as a result of perception right and that imagination and internal generation of states has a particular role to play in learning and things like this so i'm kind of rambling but um my my sense is that this um this sort of bayesian helmholtzian revolution or whatever you want to call in recent years has shown that we do understand many principles of how the brain functions at the level of like at the human scale um if the question is about like how to how do other scales fold into this i um i mean there's some work that maxwell ramstead and casper has been people like that have done on understanding their on all population dynamics in terms of active inference at that scale and so that's really interesting um uh so like what i would think of as a structural like a structural representation from the point of view of me or you like a human being um you can also think of as a process of neurons performing active inference right at their own scale um but i think there's to me the hard question there is how do you relate uh the contents of the representation at the higher scale to the contents of the lower scale that that's that's the thing that bothers me about in this area yeah just if i could just respond really quick yeah that's exactly the the thing i mean we can see how this active inference is happening at different scales for sure but how scaling up and scaling down or or the the translation from one scale to another happens is very like mysterious still yeah totally cool so we have about 20 minutes left so if anyone has a question in the live chat otherwise we'll go to scott and then stephen and be preparing our thoughts for what to continue discussing in point two so scott so um picking up in the last couple of points really interesting the i was thinking of what was the gentleman's name who barrettson who said information is the difference that makes a difference what was that guy's name is it anyway the difference that makes a difference and i always thought the first difference that he's talking about is the very um perceptual a perceptual change in the environment and the second difference is is that perceptual change relevant to you and so i want to unpack that a little bit here because contents one of the things that i was talking about my mit presentation and one of the things i noted is shannon tells us signal versus noise but it it i think but it doesn't tell us if the signal is a cancer cure or a chocolate chip cookie recipe right and so the content is left undefined there now what if so so i define living systems as auto catalytic entropy secreting structures notice i don't say they have to be embodied in any particular embodiment my son said the only reason i came up with the definition is i wanted to use the word secretion because it was juicy but he's right in part i wanted to make it juicy but the reason i raised that here is when you look at barrett-center or whoever was it barons in any way the difference that makes a difference um info information really maybe is what we

cognitive beings call relevant differentials that we perceive irrelevant differentials are just noise and so the difference that makes a difference we call it information but there's a lot of differentials out there in the world and that's the reason i'm going there is since we're talking to a philosopher today and what i deal with as a lawyer is rhetoric and rhetoric is what you do to bring together incommensurables right and rhetoric is not like crazy glue it's like it's like um epoxy it doesn't just bring things together it also brings them together and fills gaps so the rhetoric of information is our storytelling about relevant differentials and we seek to sculpt our internal model and then train other people's internal models because when we entrain other people's internal models with our rhetoric then it de-risks our externality because other people's internal models are my externality right so the rhetoric is the tool that we use to reduce differentials or to increase relevance alignment among cognitive systems running cognition how's that i wanted to try to integrate it that seemed to me that last couple of comments it felt like it may be information is an article when we put much in seeking information maybe information the difference that makes a difference after the fact we can tell that because before the fact we can't tell what's relevant for instance suntan lotion never existed because we didn't know that photons caused skin cancer now there's an entire commercial factories and areas in the grocery store that sell suntan lotion now it doesn't mean that the sun became carcinogenic we just learned that it did so the inference we were able to make that inference at large scales and now influence buying behavior which leverages and de-risks future behaviors which is the reason that we care about information that's the difference that makes a difference anyways a couple of thoughts on this last couple of things that's that's really cool stuff i mean i i really see things similarly i think on a lot of fronts uh so the question of it sounded to me like you're kind of saying like we don't have to be too fixated on this term information because it really it means differences that make a difference or being aware of differentials so i mean i i agree with that i think um i mean i i again i always go back to just the the sort of bedrock stuff so like i think i think shannon's interpretation of information was actually a little bit unnecessarily epistemic i think it could have been just doxastic so he says you know it's coming to know something it's like well but we could generalize that and just talk about what one comes to believe and then i think if we have if we work with that notion which isn't information anymore as shannon defined it because he talked about knowledge but whatever that generalization of information is called um i think that would really um conduct the logic thanks um yeah so um so that's that's all that i think we're getting at with the concept of information but what what i think it's really interesting and

difficult philosophically is okay so obviously there's there's self information right so there's just like something pings my retina and or you something downstream from that and i notice a little flip in my visual field and that's that's something i'm aware of so that's information about itself anyway but then the you know the more important signals carry all sorts of information about other things meaning they i guess they lead us to predict and expect other differences beyond themselves and so and so the tricky thing to me is okay so i take myself to be this system of interacting parts neurons and also other things obviously that um sort of embodies a perspective and the information that a signal carries depends on that whole system and how this signal impacts it right and and you know one of my neurons is a entirely different type of system here's my daughter tell me something what's up you got ice cream awesome sorry um that's awesome um yeah so your generative model's on a live stream right now i'm sorry that's cool yeah um i don't know so i was saying something about neurons and then now i'm thinking about ice cream um basically yeah so so the tr like that the question for me is so like i think of content right the information content of a signal for a person as relative to the person to the system and so if you've got a completely different system with a different structure i think you can sort of compare the structures of the systems to see how similar the content is right but i think it has to be at that whole level of the sort of holistic entire system level that you need to make these content comparisons um so i don't remember whether that directly links back to where we started this but um nice in one one quick thought on that is the the difference that makes a difference which is also a big key term in the philosophy of biology like the mutation is the difference that makes a difference for something um in active inference it's the difference in observation that makes a difference for either the generative model updating that's inference or the difference that makes a difference for action so we have a way to talk about it and then also um the doxastic note not my total area but it looks like this is how we can work towards pluralism across perspectives because if it's framed as one person believes that it's hot and one person believes that it's cold if there's only one truth and we're gonna play tug of war then we have to fight but if it's true that one person believes it's hot and it's true that one person believes it's cold there's no problem there still might be differences on what temperature the people want the room but by framing things about what a perspective or a system or an agent believes to be true or just believes period could be an ironic belief all of a sudden we can interact with multi-agent systems and deal with them in a way that still has local logic so steven and then anyone else so thanks just so closing out i suppose in a way i'd be curious how this this work with beliefs that has a nice as as daniel said it bridges action

orientated approaches and pragmatic approaches with more epistemic approaches because beliefs tend to be beliefs about what will happen when something is done often not necessarily and i'm curious how this paper and this kind of underpinning relates or helped to inform where you were going with a new paper that was intriguing about the folk psychology and how maybe that that there's some sort of that's where one of the bridges out from this paper into you know folk psychology being how we kind of intuitively believe the world is based on our experiences so there is a certain element of like whether that's how it's structured in the brain it's kind of how we might structure it before science tells us something else in a in a certain socio-cultural sense so i'd be interested in how that extension plays out with this paper yeah so so there's definitely a direct connection between this paper and and that one i mean i wanted to write a sort of sequel to this paper where i talked about specifically desires and how those the the sort of the way those would be grounded in thermodynamics according to this picture or co-native states in general like the states with a world to mind direction of fit um and so then i started talking to to ryan smith and others about this and we ended up you know co-authoring a quite different paper that has some commonality but i still want to write um i still want to write one that's more on the thermodynamic side of things but um the connection is in this you know the sort of identity thesis that at the end of this paper i argue for it by way of david lewis who who one of the terms in the identity is the implicit definition that folk psychology gives us of mental um so we haven't really gotten into that maybe we will next week or something i don't know there's ramsey sentences i'd love to talk about them we got we got the ramsey sentences we got the sleep wake simulation yeah other topics that we will really look forward to exploring yeah let's please get into that but um basically i mean i don't know if as usual i feel i always feel like i'm not doing an adequate job of answering questions i'm sorry but um but by folk psychology in this context i'm really thinking of just our our it's sort of like theory of mind right it's like our implicit theory of the way people including ourselves um function so i think that that's essential even though it's not highly until the end that's essential to what i'm doing in this paper because if you don't have mental states to define then you don't have any identity relation because that's one of the terms in the identity relation although you could also i guess maybe if you're doing pure bayesian modeling you could just write down a bunch of states and equations for how they update um without thinking about spoke psychology at all but but i guess the point of this is um if one of the terms in this relation is the mind well i think folk psychology i don't actually love that term i mean it's okay it's a philosopher's term of art right i just mean intuitive human psychology i think is one of the terms in this relation cool

scientists are folk too so cognitive diversity means that we have all different ways of knowing and there may not be one default or simply intuitive way of working or knowing because what's intuitive to one person will be out of the groove for somebody else and that's why the movement towards pluralism in terms of our understanding of ideas and of ecosystems is so interesting to talk about and see connected to active inference any um thoughts or what would be something that somebody's excited from the conversation today or thinking about how we could resolve or at least introduce again next week i think a few of the things we have written down is the sleep wake simulation what is that agent doing what does it do in terms of your model we have the ramsay sentences and where does that style of logical representation come into play we have empirical validity what kind of experiments are you interested in or curious about what what else could be interesting uh we could still talk about the the contrast of divergence versus simulated annealing thing um i just really briefly wanted to follow up and say i think when i say intuitive psychology i mean i don't mean to suggest that there's just one of those i think that but i do think we can sort of average over different perspectives to get that so like uh the individual perspectives ultimately are what's real but if we want to talk about things at an abstract level we could average over those and i think that's kind of what lewis tries to do so maybe this is something we'll get into next week um but really he when he talks about folk psychology he he he engineers it so that it doesn't have to be that there's just one set of state statements that embodies what people think about psychology it's like a disjunction of different um sets that mostly overlap um um and just yeah i mean pluralism is really i think is um is a good idea in many contexts um i actually think this would be interesting this might be controversial i think it's not always the goal in science because if you start out with pluralism then you just don't do anything right it's like okay everything's fine as it is but i think sometimes we're seeking a unifying underlying explanation and if you accept pluralism at the start then you won't get there that tension which is okay pluralism everybody must be pluralist and this is the only way it's actually absolutist and singular at a higher level and so pluralism means that we each do have our perspectives and yet there is a way of working it's not just the same thing as saying we have an absolute answer and it's that there's no answer that's the kind of local logical paradoxes that we see all around us that's a great point stephen and then yes that's that's the um the rabbit hole of post-modernism oh we don't want to go but it's it's it's good this is a really rich conversation and uh i think like you say it's uh to be able to jump from helmholtz to some of these trans-cultural things is uh you know it shows you that it's pretty foundational some of the stuff we're touching so that's great so thanks for a lovely a

lovely talk today i really enjoyed it thank you yeah thanks a lot any closing notes there alex otherwise we're all gonna probably re-listen and re-read and think about what would be good to ask next week yeah i know thank thank you all it's it's really fascinating and fun and i just want to do more of it next week i'm i'm very uh i'm i didn't have anything else to say but i kept speaking i'm not sure why same way do that do that sometimes it happens so thanks again everybody who's participating live and on live stream and we will see you next week